

A

B

C

D

1

2

3

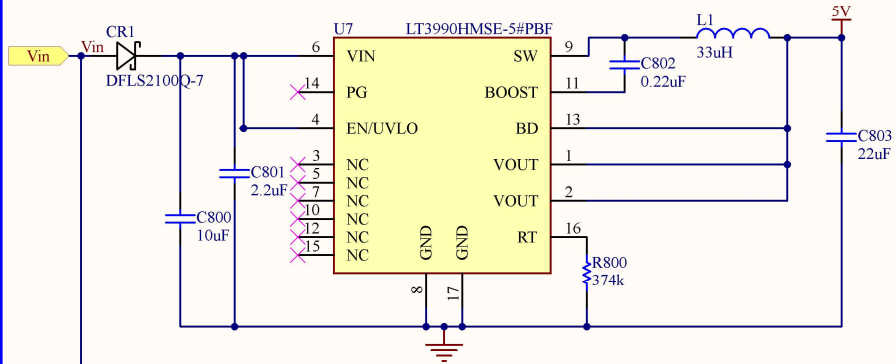
4

△ Only for EPC23102 (Transistors).
Uses only 16mA according to the datasheet.
Simulations show a initial peak around 50mA and a
triangle wave form with min 16mA and max 24mA.

EPC23102: 4.75 V (Min) 5.5 V (MAX)

5V Supply

Max Current Output 350 mA



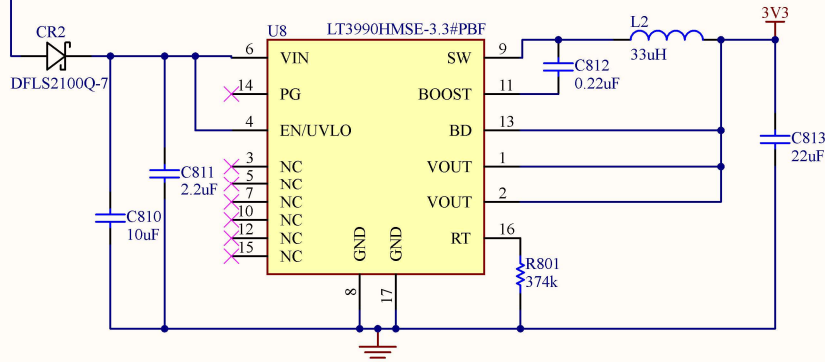
△ C1 and C2 must be rated for 100 V

C3 must be rated for 16 V

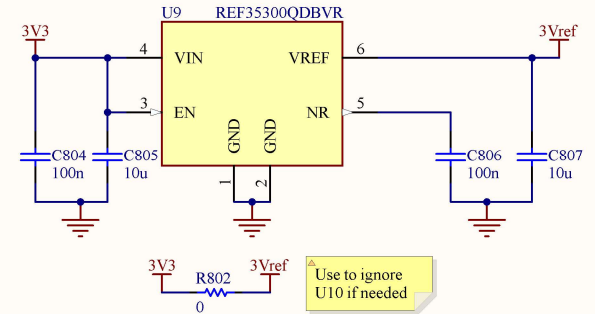
C4 must be rated for 10 V

3.3V Supply

Max Current Output 350 mA

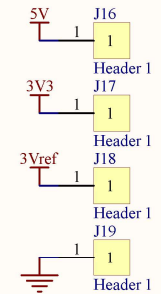


3V voltage reference



△ Use to ignore
U10 if needed

Debug Probes



△ For evrythig:
2*INA253: 1.8 mA (Typical) 2.6 mA (Max)
CAN_transceiver: 10 mA (Typical) 17 mA (Max)
STM: 11 mA (Typical) ~20/30 mA (Max)
2*AD8605: 1 mA (tyoical) 1.4 mA (Max)
Total: 26.6 mA (Typical) 55 mA (Worst case)

Title: Power Supply

Project: MPPT V1.2 Afonso.PrjPcb Version: 1

Date: 9/26/2026 Author: Afonso Coelho

Company: Técnico Solar Boat

File: PowerSupply.SchDoc

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